

Ming Gong, Ph.D.

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SUMMARY

Hands-on **applied-AI engineering leader** and repeat **0→1 builder** who forms teams from scratch, sets the technical bar, and ships production AI into enterprise environments. Currently Head of ML Engineering at Atlassian, the single point of accountability for an extended org of **130+** (60+ direct reports plus dotted-line partner teams) across Jira Search & Agent, Knowledge Fundamentals, and Generative AI Search Experiences. Previously Group Senior Principal Applied Sciences Manager at Microsoft, built and led a 40+ team that took Bing Question-Answering and Copilot Search from English-only to 100+ languages and 230+ markets, and shipped **Copilot Search as a Bing AI vertical one month ahead of Google's AI Mode**.

Operates as a **player-coach** — close enough to review architectures, debug production issues, and prototype alongside the team, with primary impact through the people hired, the standards set, and the culture created. Turns one-off wins into repeatable playbooks, templates, and frameworks. Uniquely multi-dimensional — **applied sciences & full-stack engineering, AI products & research** (50+ publications, 11K+ citations, 20+ patents), **consumer & enterprise products**. 13+ years engineering, 7+ years engineering management, 6 years prior as an IC engineer / tech lead.

SELECTED HIGHLIGHTS — ALIGNED TO FORWARD DEPLOYED ENGINEERING LEADERSHIP

- **Founding 0→1 builder.** Built Bing Question-Answering from zero; built a 40+ team at Microsoft from scratch; **Enterprise Generative AI Search** — identified a cross-product gap no team was chartered to own, POC in <1 month, GA early 2026; **Jira Search MVP in 4 months** → 3.4× Search MAU; established greenfield Canada and Singapore sites.
- **Hires, develops, and retains top engineers.** Single point of accountability for an **extended org of 130+**; grew the direct org **30 → 60+ in one year**, recruiting 2 Principal-level Managers and 3 Principal-level Tech Leads; regular 1:1s, technical mentorship, and career growth; founded **Microsoft AI School** (10K+ participants, 2K+ honored graduates).
- **Player-coach at the technical bar.** Reviews architectures and debugs alongside the team — diagnosed entity-linking failure modes through **2 months of hands-on diagnosis** to drive precision from 40% to 90%+; held a contested technical framework through 5–6 leadership reviews.
- **Ships production AI into enterprise environments.** **Atlassian Enterprise Generative AI Search, Jira Agents in Rovo Chat, Copilot Search, and Knowledge Card** — production AI delivered to enterprise and consumer customers at scale.
- **Turns one-off wins into repeatable patterns.** **Lumina** adopted as Atlassian's standard generative-UI framework across all GenAI products; reusable frameworks **Glow, Orca, and NeuronBlocks**; a Coalition Council for repeatable cross-org alignment; Microsoft AI School as a scaled enablement program.
- **Cross-functional partnership & executive presence.** V-team of ~30 → **115 contributors across 8 partner orgs**; partners across research, product, and engineering; moves fluidly from leadership reviews to hands-on debugging; live demo at **Team '26 Founder Keynote**.
- **Research depth that ships.** Best Paper Award, IJCAI '24 AI4Research; **2× Microsoft MLADS Distinguished Contribution Awards** (first-ever back-to-back; CTO Office global awards); **50+ peer-reviewed papers, 11K+ citations, 20+ granted patents**.

EXPERIENCE

Atlassian (US)

Feb 2025 – Present

Head of Machine Learning Engineering — Knowledge AI

Bellevue, WA

- Single point of accountability for the Knowledge AI charter, leading an **extended org of 130+** — 60+ direct reports (70% scientists, 30% engineers; full-stack and globally distributed) plus dotted-line partner teams across Jira Search & Agent, Knowledge Fundamentals, and Generative AI Search. Grew the team from 30 to 60+ in one year, recruiting **2 Principal-level Managers and 3 Principal-level Tech Leads**; established new teams in **Canada and Singapore** (greenfield sites for the company). [[Hire, develop, retain](#)]
- **Shipped Jira Search MVP within 4 months** of joining, full version launch to all customers within 12 months; rebuilt ranking stack and search UX, driving **3.4× Search MAU growth (2.5M → 8.5M)**, helping AI Org exceed Search MAU L1 OKR; premiered at Team '25. Formed a **V-Team growing from ~30 to 115 contributors across 8 partner orgs** with a Coalition Council to drive cross-org alignment. [[0→1 delivery](#)]
- Generative Answers in search and AI Definitions drove sustained **AI MAU from 100K to 1.2M** via a novel generative UI framework. [[Production AI shipped](#)]
- Built core **knowledge fundamentals** (collaboration graph, entity linking, entity recommendation) now adopted by multiple product teams for personalized features. Led **entity linking turnaround from 40% to 90%+ precision** — invested 2 months in hands-on diagnosis before acting, protected team stability through parallel senior hiring, then redirected technical approach. [[Player-coach / hands-on](#)]

- Launched **Jira Agents in RoVo Chat** within 3 months, becoming a primary driver of RoVo adoption. Code-driven agent architecture **beat Manus and GenSpark on the GAIA benchmark** in early 2026. Pioneered AI-native team operation by building **X-SuperAgent** (Atlassian's first OpenClaw-compliant agent framework) — boosting developer productivity by **40%+**. [\[Agentic solutions / reusable tooling\]](#)
- Initiated and led **Enterprise Generative AI Search** from zero to GA — identified a cross-product platform gap no team was chartered to own; POC in <1 month, MVP shipped in early 2026. The novel **Generative UI layer (Lumina)** — **20% engagement uplift, 30% SBS lift** vs. competitor AI experiences — was adopted as Atlassian's standard generative presentation framework, now being integrated across all GenAI products and features. Previewed at **Team '25 EU AI Super Session**; live demo at **Team '26 Founder Keynote**. [\[0→1 founding\]](#)

Microsoft — Search Technology Center Asia (Bing, Edge, Copilot)

Aug 2013 – Dec 2024

Group Senior Principal Applied Sciences Manager (2019–2024) · Senior Dev Manager (2018–2019) · Senior Tech Lead (2016–2018) · SWE II (2013–2015)

Beijing /
Suzhou

- **Built and led an org of 40+ engineers and applied scientists from scratch** (70% scientists, 30% engineers; **5 managers** — 3 science, 2 engineering) across two sites (Beijing and Suzhou), spanning Bing Search, Copilot, Cross-Lingual QA, Knowledge Graph, Recommendation, and Edge browser AI. [\[Team building 0→1\]](#)
- **Copilot Search (Bing AI vertical, 2024–2025)**: delivered multi-turn AI search experience unifying GenQnA, GenSERP, and Magazine; built scalable LLM generation pipelines and task-specific SLMs (incl. R1-distilled). **Shipped April 2025 — one month before Google's AI Mode**. [\[Ship velocity\]](#)
- **Generative SERP (2023–2024)**: reinvented Bing SERP as "AI Masterpieces" — proposed tier-wise hallucination control framework **against split organizational consensus** (conservative vs. velocity camps); held the position through 5–6 leadership reviews over two months. Framework adopted by Edge Copilot and Office Copilot. Built Glow, a **distilled SLM framework replacing GPT-4-class models** for production serving — first-token latency from 20–30s down to **~5ms**; V2 pipeline scaled to 5–10% query coverage. [\[Reusable frameworks\]](#)
- **Generative Answers on Bing SERP (2023–2024)**: **20% query coverage on Bing global markets, +2.0% WPSBS** via streaming generative answers and click-through to Bing Chat. [\[Production delivery\]](#)
- **Knowledge Card for 150M Entities (2021–2023)**: multi-modal entity aggregation across web sources with template + MRC DL extraction and LLM generation. **60% Bing query traffic, 65M DAU, +3.0% WPSBS**; certified as a Bing "WoW" feature (<10 out of 100+ features). [\[Customer-scale impact\]](#)
- **Bing Question-Answering: Zero-to-Global, Traditional ML → DL → Pre-trained Models → LLM (2016–2024)**: built Bing QnA from the ground up and led its evolution through every major AI paradigm shift. Placed the universal multilingual bet **~1 year before globalization became an official org strategy**. Scaled from English-only to **100+ languages and 230+ markets** via systematic cross-lingual methodology — **2× MLADS Distinguished Contribution Award** (first-ever back-to-back recipient; CTO Office global awards). Shipped 10+ DNN releases that replaced legacy GBDT ranker (400+ hand-crafted features) and won WPSBS vs. Google. Co-owned NRT inference pipeline design with **8 platform teams across time zones**, treating partners as co-owners with shared credit — pipeline became Platform team's flagship demo and blueprint for subsequent products. Authored novel model compression that powered the **first BERT model deployed in Bing** (within 1 month of BERT open-source, zero additional GPU cost) — **+3.0% coverage, +5.0% precision**. [\[0→global / repeatable patterns\]](#)
- **Query-Dependent Passage Extraction (2019–2022)**: replaced industry-standard sliding-window passage indexing with runtime query-dependent extraction via GNN document modeling + multi-teacher distillation to 40ms latency. **Held #1 on Google Natural Questions benchmark for ~1 year**; architecture adopted across Bing QnA, Knowledge, People Also Ask, and Azure Custom QnA. [\[Hands-on technical depth\]](#)
- **Segment Entity Search & Recommendation (2021–2024)**: shipped recipe, job, and movie answers — **beat Google on relevance and WPSBS** on recipe and job; closed WPSBS gap with Google on movie recommendation. [\[Quality bar\]](#)
- **Microsoft Copilot on Edge (2023–2024)**: long-document summarization and chat over PDFs and long-form web pages — 1000+ page docs at **95%+ accuracy**; demoed at Microsoft Copilot 2023 announcement event; shipped to Edge and Windows Copilot. [\[New solution surface\]](#)
- **Underside on Edge — Right-Rail Insight Pane (2022–2024)**: built the AI-powered right-rail panel for Edge featuring page summary, topics, QnA, related search, and related pages (feeds). **Edge engaged DAU +10%, Search DAU +1.2M**. [\[Customer impact\]](#)
- **NLU Platform "Orca" (2020–2025)**: unified query understanding pipeline (domain, intent, slots, topics) powering answer triggering. **95%+ of Bing answers (coverage ≥ 0.05%) onboarded**; 60% of onboarding answers closed triggering gap with Google. [\[Reusable platform\]](#)

PLATFORMS & INTERNAL TOOLING BUILT

- **Lumina** (Atlassian, 2025–2026) — generative UI framework powering Enterprise GenAI Search; **20% engagement uplift, 30% SBS lift** vs. competitor AI experiences; adopted as Atlassian's standard generative presentation framework across all GenAI products.
- **X-SuperAgent** (Atlassian, 2025–2026) — Atlassian's first OpenClaw-compliant agent framework for AI-native team operation; **40%+ developer-productivity uplift**.

- **Glow** (Microsoft, 2023–2024) — distilled SLM serving framework replacing GPT-4-class models for production; first-token latency **20–30s → ~5ms**; V2 scaled to 5–10% Bing query coverage.
- **Orca** (Microsoft, 2020–2025) — unified NLU platform (domain, intent, slots, topics) powering answer triggering for **95%+ of Bing answers**.
- **NeuronBlocks** (Microsoft, 2019, open source) — "Build NLP DNN models like playing Lego." Published at EMNLP/IJCNLP 2019.

PEOPLE & ORG DEVELOPMENT

- **Founder & Head, Microsoft AI School — Asia-Pacific R&D Group** (2017–2024). Led a **50+ operation v-team**; drove planning, curriculum, and execution of all AI learning events for the region. **10K+ participants, 2K+ honored graduates**, many of whom grew into AI talents and leaders across the Microsoft APRD org (with strict graduation criteria); record-breaking participation 8 years running.
- **Invited instructor, LinkedIn Global Academy for Chinese Programmers** (2022–2023): 4-class series on "Intelligent QA Systems for Search Engines" — **50M impressions, 98% satisfaction**.
- **Invited speaker, InfoQ AICon Global AI Conference** (2021): represented Microsoft China on globalization of Bing QA.

EDUCATION

Ph.D., Graphics and Visual Computing — Institute of Computing Technology, Chinese Academy of Sciences

2008 – 2013

SELECTED AWARDS

- **Best Paper Award**, IJCAI '24 AI4Research — "*Step-Back Profiling: Distilling User History for Personalized Scientific Writing.*"
- **2021 Fall Microsoft MLADS Distinguished Contribution Award** (2 / 250+). Broke conference history to receive the award twice consecutively.
- **2021 Spring Microsoft MLADS Distinguished Contribution Award** (2 / 300+).
- 2023 Microsoft Global Hackathon — Edge Theatre, Hack for Web Experiences: **Best Overall Hack**; Greater China Region (Beijing): Most Popular Hack.
- 2023 Microsoft Global Hackathon — WebXT China, DeKG (decentralized knowledge ecosystem): Best Innovativeness & Originality Hack; Beijing Outstanding Project Award.
- Excellent Operation Award of AI School China — 2018, 2019, 2020, 2021, 2022, 2023.
- Microsoft AI Core Q3 FY18 Greatness Award (2018).

SELECTED PUBLICATIONS (50+ papers, 11,000+ citations on Google Scholar — full list on [scholar profile](#))

► LLM, Agentic AI & Multimodal Pre-training

- Yaobo Liang, Chenfei Wu, Ting Song, et al., **Ming Gong**, Nan Duan. "TaskMatrix.AI: Completing Tasks by Connecting Foundation Models with Millions of APIs." *Intelligent Computing* 2023.
- Gen Li, Nan Duan, Yuejian Fang, **Ming Gong**, Daxin Jiang. "Unicoder-VL: A Universal Encoder for Vision and Language by Cross-modal Pre-training." *AAAI* 2019.
- Nuo Chen, Ning Wu, Shining Liang, **Ming Gong**, et al. "Is Bigger and Deeper Always Better? Probing LLaMA Across Scales and Layers." *arXiv* 2023.
- Ning Wu, **Ming Gong**, Linjun Shou, et al. "Large Language Models are Diverse Role-Players for Summarization Evaluation." *NLPCC* 2023.

► Model Training, Compression & Distillation

- **Ming Gong**, Linjun Shou, Wutao Lin, et al. "NeuronBlocks — Building Your NLP DNN Models Like Playing Lego." *EMNLP/IJCNLP* 2019.
- Ze Yang, Linjun Shou, **Ming Gong**, Wutao Lin, Daxin Jiang. "Model Compression with Two-stage Multi-teacher Knowledge Distillation for Web Question Answering System." *WSDM* 2020.
- Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, **Ming Gong**, Daxin Jiang. "Reinforced Multi-Teacher Selection for Knowledge Distillation." *AAAI* 2021.

► Code Intelligence

- Zhangyin Feng, Daya Guo, Duyu Tang, Nan Duan, Xiaocheng Feng, **Ming Gong**, Linjun Shou, Bing Qin, Ting Liu, Daxin Jiang, Ming Zhou. "CodeBERT: A Pre-Trained Model for Programming and Natural Languages." *EMNLP (Findings)* 2020. (3,000+ citations)
- Junjie Huang, Duyu Tang, Linjun Shou, **Ming Gong**, et al. "CoSQA: 20,000+ Web Queries for Code Search and Question Answering." *ACL* 2021.

► Cross-Lingual & Multilingual Modeling

- H. Huang, Yaobo Liang, N. Duan, **Ming Gong**, et al. "Unicoder: A Universal Language Encoder by Pre-training with Multiple Cross-lingual Tasks." *EMNLP/IJCNLP* 2019.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei, Daxin Jiang. "Bridging the Gap between Language Models and Cross-Lingual Sequence Labeling." *NAACL* 2022.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei. "From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension." *AAAI* 2022.

- Shining Liang, **Ming Gong**, Jian Pei, Linjun Shou, Daxin Jiang. "Reinforced Iterative Knowledge Distillation for Cross-Lingual Named Entity Recognition." *KDD* 2021.
- Nuo Chen, Linjun Shou, Tengtao Song, **Ming Gong**, et al. "Structural Contrastive Pretraining for Cross-Lingual Comprehension." *ACL* 2023.
- Linjun Shou, **Ming Gong**, Jian Pei, et al. "Language Scaling: Applications, Challenges and Approaches." *KDD Tutorial* 2021.
- Linjun Shou, **Ming Gong**, Jian Pei, et al. "Scaling out NLP Applications to 100+ Languages." *WWW Tutorial* 2021.

► Dense Retrieval & RAG

- Shengyao Zhuang, Linjun Shou, Jian Pei, **Ming Gong**, et al. "Typos-aware Bottlenecked Pre-Training for Robust Dense Retrieval." *SIGIR* 2023.
- Houxing Ren, Linjun Shou, Jian Pei, Ning Wu, **Ming Gong**, Daxin Jiang. "Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval." *EMNLP* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Modeling Sequential Sentence Relation to Improve Cross-lingual Dense Retrieval." *ICLR* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Multi-View Document Representation Learning for Open-Domain Dense Retrieval." *ACL* 2022.
- Linjun Shou, S. Bo, Fei-Xiang Cheng, **Ming Gong**, et al. "Mining Implicit Relevance Feedback from User Behavior for Web Question Answering." *KDD* 2020.

► Knowledge & Entity Understanding

- Zilin Xiao, **Ming Gong**, Jie Wu, et al. "Instructed Language Models with Retrievers Are Powerful Entity Linkers." *EMNLP* 2023.
- Zilin Xiao, Linjun Shou, Xingyao Zhang, Jie Wu, **Ming Gong**, et al. "Coherent Entity Disambiguation via Modeling Topic and Categorical Dependency." *Findings of EMNLP* 2023.

► Recommendation & Personalization

- Ning Wu, **Ming Gong**, Linjun Shou, Jian Pei, Daxin Jiang. "RUEL: Retrieval-Augmented User Representation with Edge Browser Logs for Sequential Recommendation." *CIKM* 2023.
- Hanshuang Tong, Jun Li, Ning Wu, **Ming Gong**, et al. "Plutos: Towards Interpretable Stock Movement Prediction with Financial Large Language Model." *arXiv* 2024.

SELECTED PATENTS (20+ granted; selection below)

- Model Generation based Model Compression. 2019. (MS#406805-CN-NP)
- Model Selection Learning for Knowledge Distillation. 2020. (MS#408426-CN-NP)
- Multi-Model Joint Denoising Training. 2021. (MS#409566-CN-NP)
- Contrastive Learning for Question Answering. 2020. (MS#407988-CN-NP)
- Providing QA Training Data and Training a QA Model based on Implicit Relevance Feedbacks. 2020. (MS#407927-CN-NP)
- Question Generation from Queries. 2021. (MS#410241-CN-NP)
- Assertion-based Question Answering with Question-Aware Open Information Extraction. 2019.
- Representation Learning of Semi-structured Data. 2020. (MS#408908-CN-NP)
- Graph based Fact Clustering for Diversity and Freshness. 2021.
- Controversial Poll Auto Generation by Generation Model and Crowdsourcing. 2021.
- Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval. 2022. (MS#412496-CN-NP)
- Sentence Representation Generation for Cross-lingual Retrieval. 2022.
- Representation Learning of Cross-Language Texts. 2021. (MS#409565-CN-NP)
- Cross-Lingual Task Training. 2019. (MS#406497-CN-NP)
- From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension. 2021.
- Personalized Multilingual Content Recommendation based on Robust Feature Network. 2023. (MS#412659-CN-NP)
- Sequential Recommendation based on Cross-Domain Behavior Data. 2023.
- Sequential Recommendation Based on Dual Contrastive Interest Learning. 2022.
- Temporal Co-Contrastive Learning-Based Node Representation Generation. 2022.
- Content Recommendation Based on Graph Enhanced Collaborative Filtering. 2022.
- Knowledge-enhanced Content-Aware Collaborative Filtering for Recommendation. 2021.
- Hierarchical Representation Learning of User Interest. 2021. (MS#410462-CN-NP)